Gayatri Vidya Parishad College of Engineering Autonomous

Artificial Intelligence and Machine Learning | Course Code: AI301

Machine Learning Foundations - Laboratory Manual & Practical Guide

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Syllabus Focus: AI301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Mathematical Foundations & Linear Regression

" Syllabus Scope: Linear algebra, Maximum Likelihood Estimation MLE, Ordinary least squares, Ridge L2 and

Unit 2: Classification & Probabilistic Models

" Syllabus Scope: Logistic regression, Sigmoid derivation, Cross-entropy loss, Naive Bayes classifier, Ga

AI301: Machine Learning Foundations - Advanced Topics & Assessment

Academic Track: Artificial Intelligence and Machine Learning - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Support Vector Machines & Kernel Methods

" Maximum margin classifiers, Primal and Dual optimization, Karush-Kuhn-Tucker KKT conditions, Polynomial and

Unit 4: Decision Trees & Ensemble Learning

" Information gain, Gini impurity, ID3, C4.5, CART, Bootstrap Aggregation Bagging, Random Forests, AdaBoost, G

Unit 5: Unsupervised Learning & Dimensionality Reduction

" K-Means clustering, Hierarchical clustering, Expectation-Maximization EM, Principal Component Analysis PCA e

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of Batch and Stochastic Gradient Descent for Multivariate Linear Regression from Sc

Lab 2: Binary and Multi-Class Logistic Regression Classification with Confusion Matrix Metrics

Lab 3: Hyperparameter Tuning of Support Vector Machine SVM with Linear and RBF Kernels

Lab 4: Building an Ensemble Classifier using Random Forest and XGBoost with Cross-Validation

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Christopher M. Bishop - Pattern Recognition and Machine Learning Springer

* Kevin P. Murphy - Machine Learning: A Probabilistic Perspective MIT Press

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